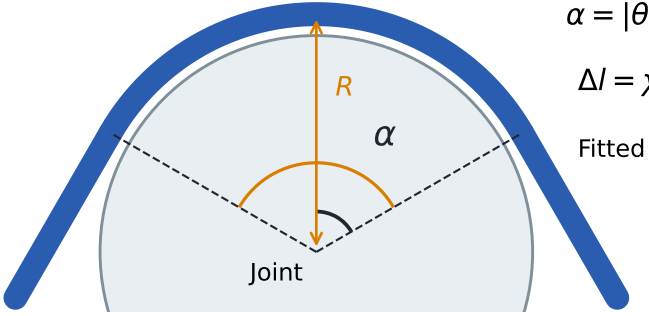
BPA following joint surface



$$\alpha = |\theta_{\text{wrap}} - \theta_k|$$

$$\Delta l = \chi_3 R \alpha \zeta^2$$

Fitted usable-length loss